SRINIVAS UNIVERSITY

INSTITUTE OF ENGINEERING AND TECHNOLOGY

MUKKA, MANGALURU-574146

(SUBJECT: ADVANCE MACHINE LEARNING)
(SUBJECT CODE:24SAL051)

A INDIVIDUAL TASK REPORT ON

"Comparative Analysis of ML Algorithms for Breast Cancer"

Submitted in the partial fulfillment of the requirements for the fifth semester

BACHELOR OF TECHNOLOGY IN AIML

Submitted by,

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UNDER THE GUIDENCE OF
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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

2026-27

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CERTIFICATE

This is to certify that **SUDESH (01SU24AI104)** has satisfactorily completed the assessment (Individual Task Report) in **Advanced Machine Learning (24SAL051)** prescribed by the Srinivas University for the 5th semester B. Tech. course during the year **2026-27**.

MARKS AWAREDED

Staff In Charge:-

Prof. Mahesh Kumar V B

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1. Introduction

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that enables computers to learn patterns from data and make accurate predictions without being explicitly programmed. Instead of relying on fixed rules, ML algorithms analyze historical data, identify relationships between features, and improve their performance through training. Today, machine learning is widely used in healthcare, finance, agriculture, education, and industrial automation to solve complex real-world problems.

One of the most important applications of machine learning is medical diagnosis, where it assists healthcare professionals in detecting diseases at an early stage. Breast cancer is one of the leading causes of cancer-related deaths among women, and early diagnosis significantly improves treatment success and patient survival. By analyzing medical data, machine learning models can accurately classify tumors as Benign (non-cancerous) or Malignant (cancerous), providing valuable support for clinical decision-making.

This project uses the Wisconsin Diagnostic Breast Cancer (WDBC) dataset to compare the performance of three supervised machine learning algorithms: Logistic Regression, Random Forest, and Support Vector Machine (SVM). The dataset contains 569 patient records with 30 numerical features extracted from Fine Needle Aspirate (FNA) images of breast cell nuclei. After preprocessing the data, each algorithm is trained and evaluated using metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

The main objective of this study is to identify the most accurate and reliable machine learning model for breast cancer prediction. The results demonstrate how machine learning can improve healthcare by supporting faster, more consistent, and data-driven diagnosis of breast cancer.

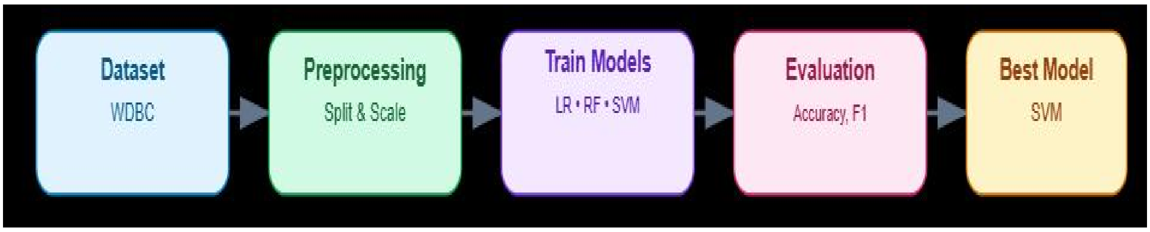
Purpose of this project

The purpose of this project is to compare the performance of three supervised machine learning algorithms using the Wisconsin Diagnostic Breast Cancer (WDBC) dataset. The selected algorithms are:

- Logistic Regression
- Random Forest Classifier
- Support Vector Machine (SVM)

Each model is trained on the same dataset and evaluated using accuracy, precision, recall, F1-score, and ROC-AUC to identify the best-performing algorithm for breast cancer prediction.

Workflow of the project



This workflow begins with loading the dataset, followed by preprocessing the data. The three machine learning algorithms are then trained, their performance is evaluated, and the model with the highest accuracy and reliability is selected as the best classifier.

2. Problem Statement

Breast cancer is one of the most common and life-threatening diseases affecting women worldwide. It occurs when abnormal cells in the breast grow uncontrollably and form a tumor. Early and accurate diagnosis is essential because it significantly increases the chances of successful treatment and improves patient survival rates. Traditional diagnosis relies on medical examinations, imaging techniques, and laboratory analysis performed by experienced specialists. However, manual diagnosis can be time-consuming and may sometimes be influenced by human error or variations in clinical interpretation.

Machine Learning (ML) provides a powerful approach for improving the accuracy and speed of disease diagnosis. By analyzing large amounts of medical data, ML algorithms can identify hidden patterns and relationships that help distinguish between benign and malignant tumors. These intelligent models support healthcare professionals by providing reliable predictions and assisting in clinical decision-making, especially during the early stages of diagnosis.

In this project, the Wisconsin Diagnostic Breast Cancer (WDBC) dataset is used to develop a breast cancer classification model. The dataset contains 569 patient records with 30 numerical features extracted from digitized Fine Needle Aspirate (FNA) images of breast cell nuclei. These features describe important characteristics such as radius, texture, perimeter, area, smoothness, concavity, and symmetry, which are valuable indicators for identifying cancerous cells.

The main objective of this study is to classify breast tumors into two categories: Benign (non-cancerous) and Malignant (cancerous). To achieve this objective, three supervised machine learning algorithms are implemented: Logistic Regression, Random Forest, and Support Vector Machine (SVM). Each model is trained using the same dataset and evaluated under identical conditions to ensure a fair comparison.

The performance of the models is measured using evaluation metrics including Accuracy, Precision, Recall, F1-Score, and ROC–AUC. These metrics help determine which algorithm provides the most accurate and reliable predictions. The final goal of the project is to identify the best-performing machine learning model that can support early breast cancer diagnosis and contribute to more effective healthcare decision-making.

3. Dataset Description

The Wisconsin Diagnostic Breast Cancer (WDBC) dataset is a widely used medical dataset for binary classification problems in machine learning. It was created using digitized Fine Needle Aspirate (FNA) images of breast masses, where microscopic images of cell nuclei were analyzed to extract important morphological features. The dataset is commonly used for developing and evaluating machine learning models for breast cancer diagnosis.

The dataset contains a total of 569 patient records, with 357 benign and 212 malignant tumor cases. Each record includes 30 numerical features that describe the characteristics of cell nuclei, such as radius, texture, perimeter, area, smoothness, compactness, concavity, symmetry, and fractal dimension. These features are measured as mean values, standard errors, and worst (largest) values, providing detailed information for accurate classification.

There are no missing or null values in the dataset, making it clean and suitable for machine learning applications. The target variable has two classes: Benign (1) representing non-cancerous tumors and Malignant (0) representing cancerous tumors. Since the output labels are already available, the dataset is ideal for supervised learning classification algorithms.

In this project, the WDBC dataset is used to train and compare three machine learning models: Logistic Regression, Random Forest, and Support Vector Machine (SVM). The dataset is divided into training and testing sets, allowing the models to learn from historical data and evaluate their prediction performance on unseen patient records. Its balanced structure, high-quality features, and medical relevance make it an excellent choice for breast cancer prediction and comparative machine learning analysis.

Dataset Summary

Attribute	Value
Total Records	569
Features	30
Benign Cases	357
Malignant Cases	212
Missing Values	0

This dataset is suitable for supervised machine learning because it contains labeled data and is widely used to evaluate classification algorithms.

4. Methodology

The methodology describes the complete process followed to build and compare the machine learning models. The dataset is first loaded and preprocessed, then divided into training and testing sets. Three algorithms—Logistic Regression, Random Forest, and Support Vector Machine (SVM)—are trained and evaluated using performance metrics to identify the best model.

Workflow Diagram

This workflow ensures a fair comparison by training all models on the same dataset and testing them using identical evaluation criteria.

Step 1: Import Libraries

```
import pandas as pd
import numpy as np

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
```

Step 2: Load Dataset

```
data = load_breast_cancer()

df = pd.DataFrame(data.data, columns=data.feature_names)
df["target"] = data.target

print(df.head())
print(df.shape)
```

Step 3: Train-Test Split

```
X = df.drop("target", axis=1)
y = df["target"]
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.20,
    stratify=y,
    random_state=42
)
```

Step 4: Feature Scaling

```
scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

5. System Architecture

The system architecture describes the complete workflow of the breast cancer prediction system, from data input to the final prediction. It shows how patient data is processed, analyzed, and classified using machine learning algorithms. A well-designed architecture ensures that the data flows through each stage in a structured and reliable manner.

The process begins with the Wisconsin Diagnostic Breast Cancer (WDBC) dataset, which contains 30 numerical features extracted from Fine Needle Aspirate (FNA) images. The data is first preprocessed by cleaning, splitting into training and testing sets, and applying feature standardization. The processed data is then provided to three supervised machine learning models: Logistic Regression, Random Forest, and Support Vector Machine (SVM).

Each model learns patterns from the training data and predicts whether a tumor is Benign or Malignant. Finally, the predictions are evaluated using performance metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC. The model with the best performance is selected as the final classifier.

Advantages of this Architecture

- Organized and easy-to-understand workflow
- Fair comparison using the same preprocessed dataset
- Supports multiple ML algorithms in one pipeline
- Produces accurate and reliable breast cancer predictions

6. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is the process of examining and understanding the dataset before building machine learning models. It helps identify the structure of the data, detect missing values or outliers, analyze feature distributions, and understand the relationship between variables. In this project, EDA is performed on the Wisconsin Diagnostic Breast Cancer (WDBC) dataset to distinguish the characteristics of Benign and Malignant tumors.

The dataset contains 569 patient records with 30 numerical features describing cell nucleus characteristics such as radius, texture, perimeter, area, smoothness, and concavity. During EDA, the dataset is checked for data quality, and it is confirmed that there are no missing or null values, making it suitable for machine learning analysis. The class distribution shows 357 benign and 212 malignant cases, indicating a moderately balanced dataset.

Several visualizations are used to better understand the data. A bar chart displays the number of benign and malignant cases, while histograms show the distribution of important features such as worst area and mean radius. A correlation heatmap identifies relationships between numerical features and helps recognize highly correlated variables that influence prediction. These analyses reveal that malignant tumors generally have larger area, perimeter, and concave points than benign tumors.

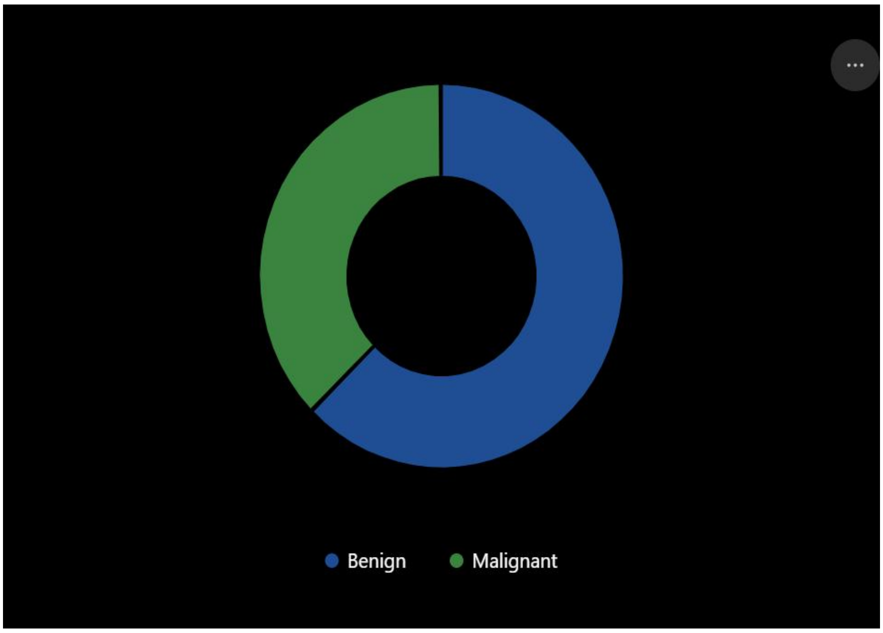
Overall, EDA provides valuable insights into the dataset and ensures that the data is clean, well-understood, and ready for model development. It forms the foundation for building accurate and reliable machine learning models for breast cancer prediction.

Main objectives of EDA

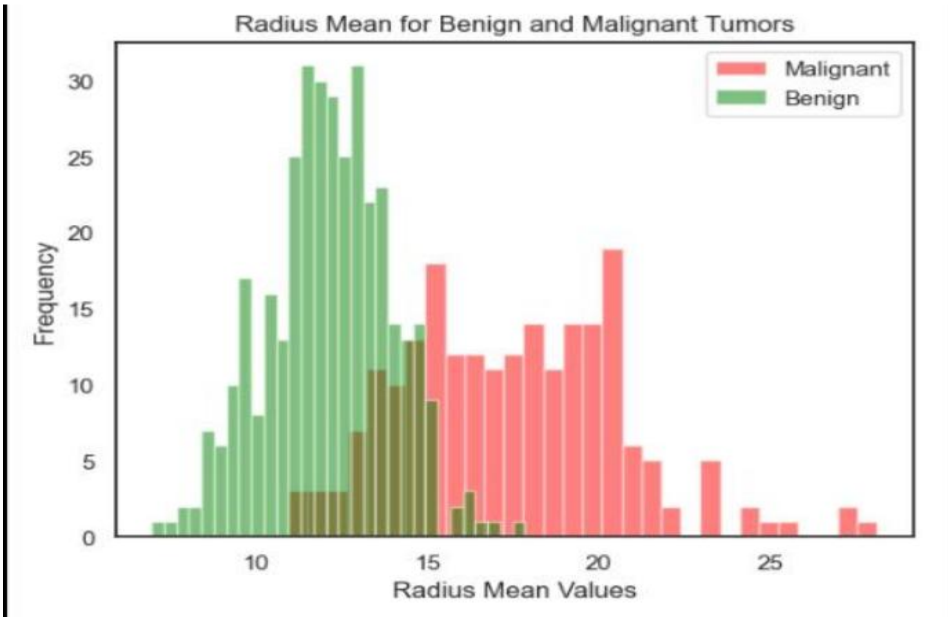
- Analyze the distribution of benign and malignant cases.
- Visualize important features using graphs.
- Identify relationships between numerical features.
- Prepare the data for accurate model training.

Graph : Class Distribution

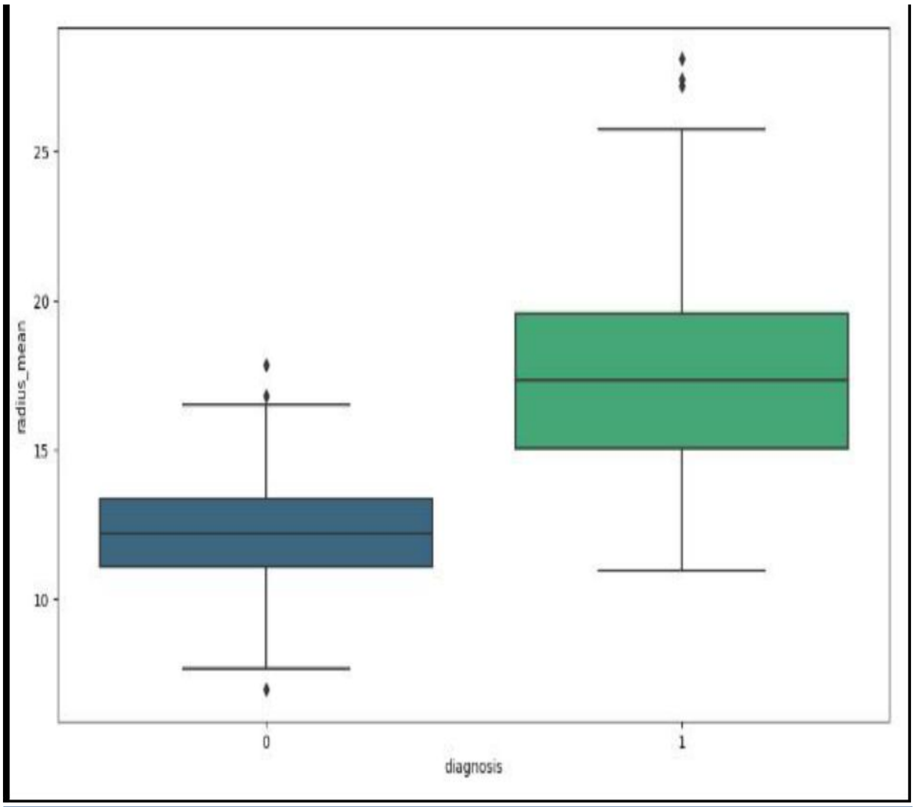
The dataset contains more benign cases than malignant cases.



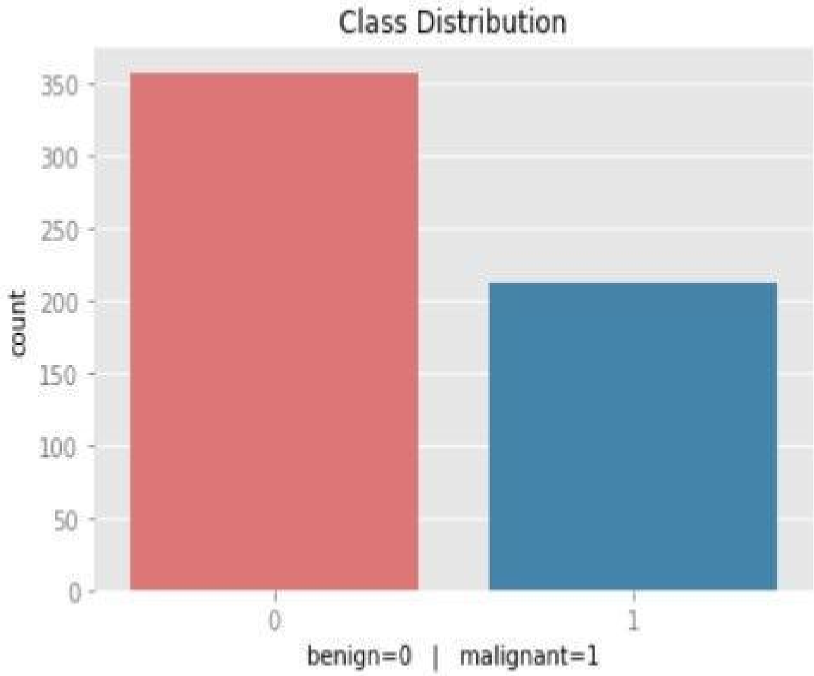
Graph : Feature Distribution




```
Most frequent malignant radius mean is: 20.1020
Most frequent benign radius mean is: 11.3286
```



Suggested graphs for this section



Class Distribution

These visualizations show that the dataset is well-structured and reveal important feature relationships that improve the performance of machine learning models.

6. Model Development

Model development is the process of building, training, and testing machine learning models using the prepared dataset. In this project, the Wisconsin Diagnostic Breast Cancer (WDBC) dataset is divided into 80% training data and 20% testing data. Before training, the numerical features are standardized using StandardScaler to improve the performance of distance-based algorithms such as Logistic Regression and Support Vector Machine (SVM).

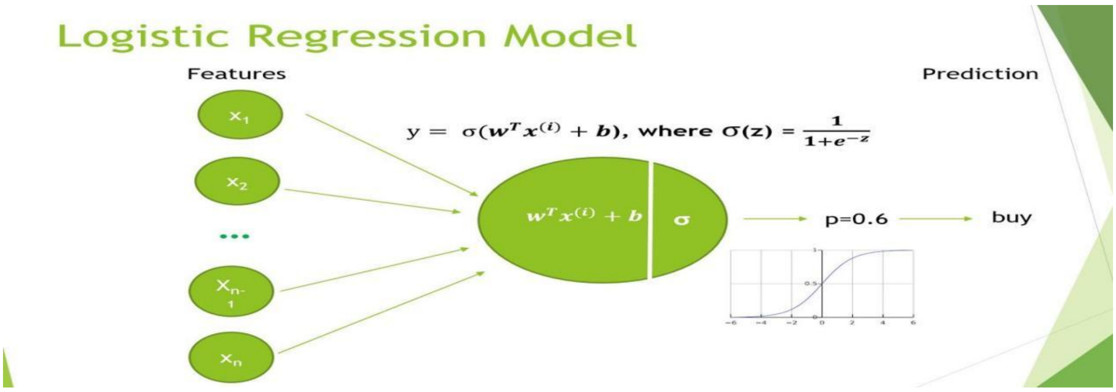
Three supervised machine learning algorithms are implemented and compared: Logistic Regression, Random Forest, and Support Vector Machine (SVM). Logistic Regression acts as the baseline linear classifier, Random Forest uses multiple decision trees to improve prediction accuracy, and SVM finds the optimal decision boundary with the maximum margin between benign and malignant tumors. Each model is trained using the same training data to ensure a fair comparison.

After training, the models are tested on unseen patient records, and their performance is evaluated using Accuracy, Precision, Recall, F1-Score, and ROC–AUC. The algorithm with the highest overall performance is selected as the best model for breast cancer prediction.

Algorithms Used

1. Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for binary classification problems. It predicts the probability that a data sample belongs to one of two classes—in this project, Benign (0) or Malignant (1). The algorithm uses the sigmoid function to convert a linear prediction into a probability value between 0 and 1. If the probability is greater than 0.5, the sample is classified as malignant; otherwise, it is classified as benign.



Code snippet

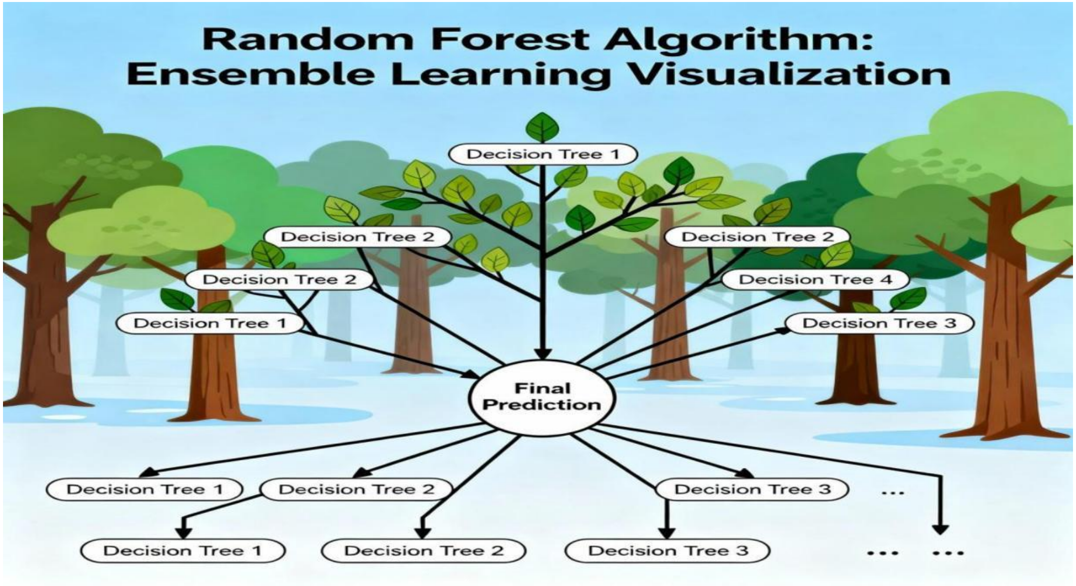
```
from sklearn.linear_model import LogisticRegression

lr = LogisticRegression(max_iter=500)

lr.fit(X_train_scaled, y_train)

lr_pred = lr.predict(X_test_scaled)
```

Advantages

- **Fast training**
- **Easy to interpret**
- **Good baseline model**

2. Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy. Each tree is trained on a random subset of the training data, and the final prediction is obtained through majority voting. This approach reduces overfitting and performs well on complex, non-linear datasets. It also provides feature importance, helping identify which variables contribute most to breast cancer prediction.



Code snippet

```
from sklearn.ensemble import RandomForestClassifier


rf = RandomForestClassifier(n_estimators=200,random_state=42)

rf.fit(X_train, y_train)

rf_pred = rf.predict(X_test)
```

Advantages

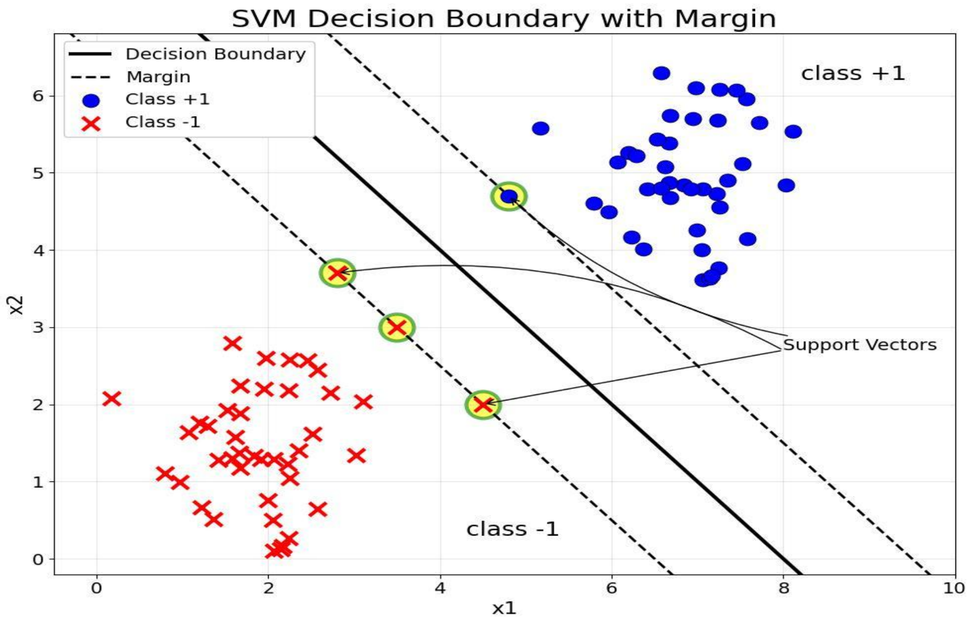
- **High accuracy**
- **Handles non-linear data**
- **Provides feature importance**

3. Support Vector Machine (SVM)

Support Vector Machine (SVM) is a powerful supervised learning algorithm used for classification tasks. It works by finding the optimal hyperplane that maximizes the margin between different classes. In this project, the Radial Basis Function (RBF) kernel is used to classify breast tumors accurately, even when the data is not linearly separable. SVM achieved the best overall performance in this study.

Code snippet

```
from sklearn.svm import SVC


svm = SVC(kernel="rbf",probability=True,random_state=42)

svm.fit(X_train_scaled, y_train)

svm_pred = svm.predict(X_test_scaled)
```



Advantages

- **Excellent for high-dimensional datasets**
- **Strong generalization ability**
- **Performs well in medical diagnosis**

Training Code Snippet

```
# Train models
lr.fit(X_train_scaled, y_train)
rf.fit(X_train, y_train)
svm.fit(X_train_scaled, y_train)
```

This section demonstrates how the three models are created and trained before evaluating their performance.

7. Evaluation Metrics

Evaluation metrics are used to measure how accurately each machine learning model predicts breast cancer. These metrics help compare the performance of Logistic Regression, Random Forest, and Support Vector Machine (SVM).

Metrics Used

- Accuracy: Percentage of correctly classified tumors.
- Precision: Measures how many predicted malignant cases are actually malignant.
- Recall (Sensitivity): Measures the model's ability to correctly identify malignant tumors.
- F1-Score: Harmonic mean of Precision and Recall, providing a balanced performance measure.

- ROC–AUC: Evaluates the model's ability to distinguish between benign and malignant classes.

Code Snippet

```
from sklearn.metrics import accuracy_score, precision_score

from sklearn.metrics import recall_score, f1_score


print("Accuracy :", accuracy_score(y_test, svm_pred))

print("Precision:", precision_score(y_test, svm_pred))

print("Recall   :", recall_score(y_test, svm_pred))

print("F1 Score :", f1_score(y_test, svm_pred))
```

These metrics are used to compare all three algorithms and identify the best-performing model.

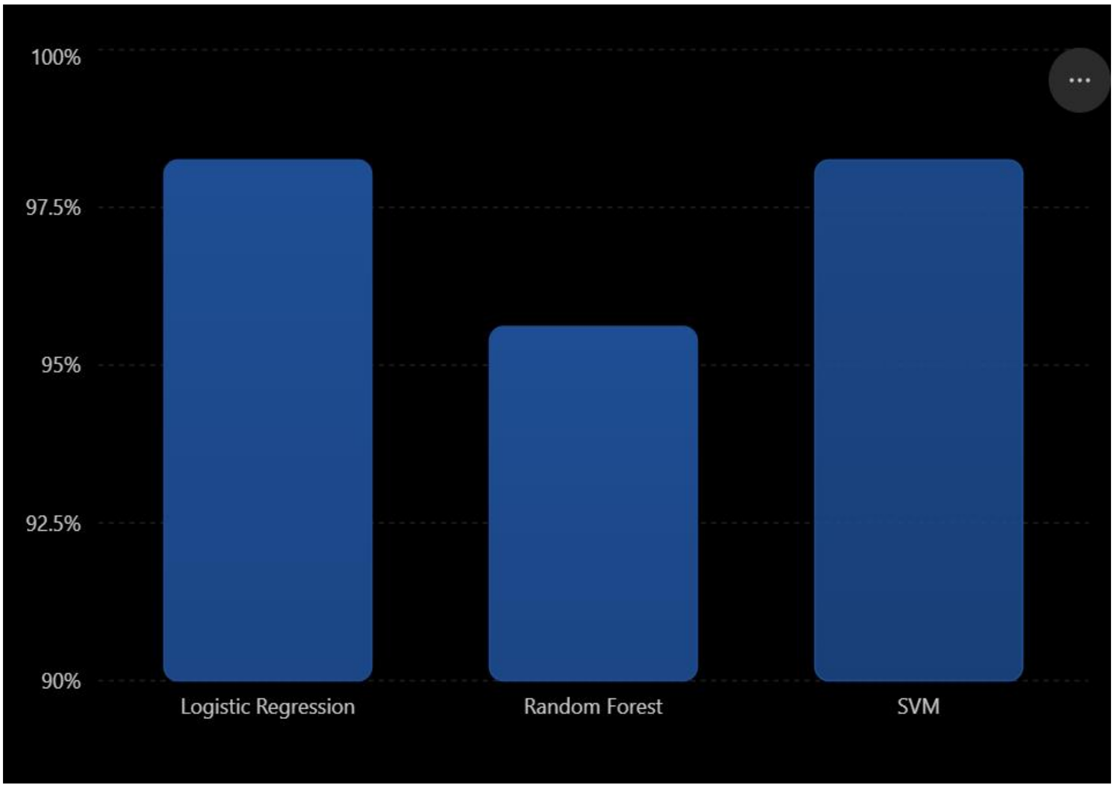
8. Results and Performance Comparison

The three machine learning algorithms were evaluated using the same testing dataset. Their performance was compared based on Accuracy, Precision, Recall, and F1-Score. The results show that Support Vector Machine (SVM) achieved the best overall performance.

Performance Comparison Table

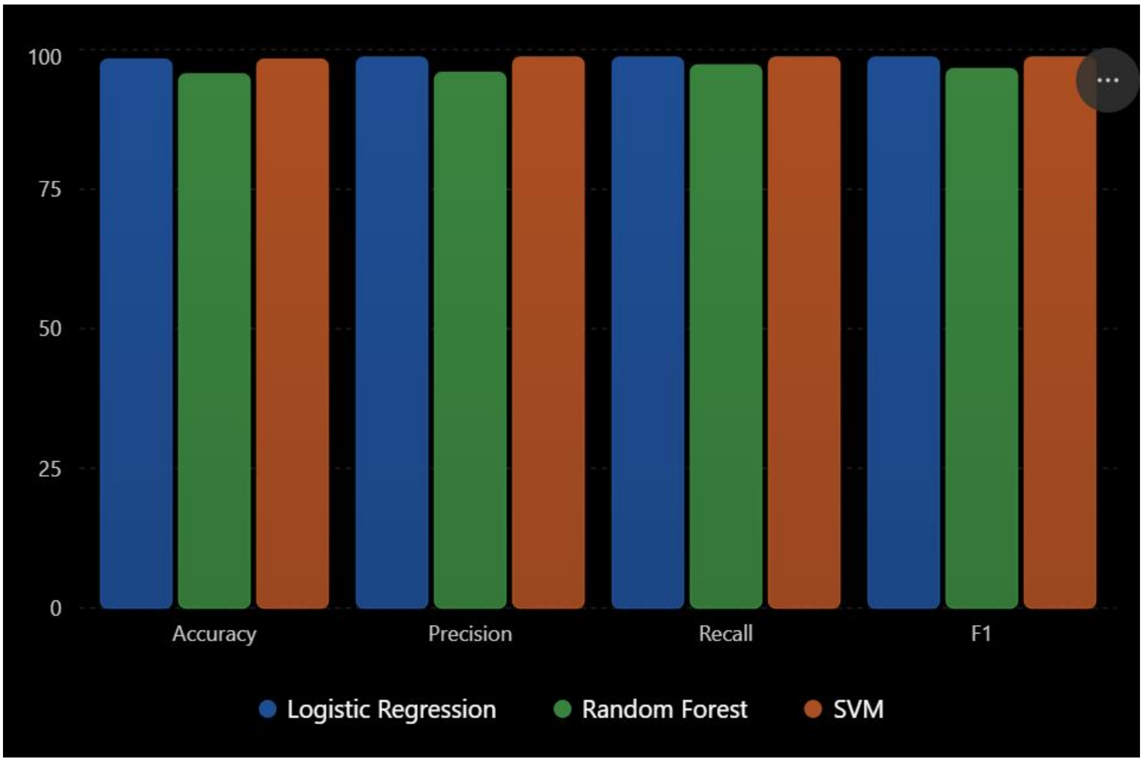
Algorithm	Accuracy	Precision	Recall	F1-Score
Logistic Regression	98.25%	98.61%	98.61%	98.61%
Random Forest	95.61%	95.89%	97.22%	96.55%
Support Vector Machine	98.25%	98.61%	98.61%	98.61%

Accuracy Comparison Graph



Observation: SVM and Logistic Regression achieved the highest accuracy (98.25%), while Random Forest showed slightly lower performance.

Graph : Multi-Metric Comparison



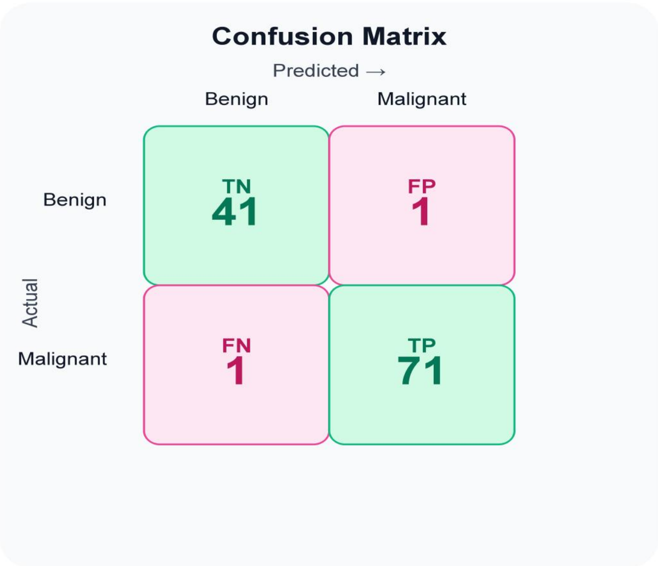
9. Confusion Matrix

A confusion matrix shows how many predictions are correct and incorrect.

Code snippet

```
from sklearn.metrics import confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt

cm = confusion_matrix(y_test, svm_pred)

sns.heatmap(cm,
            annot=True,
            fmt="d",
            cmap="Blues")

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```

Graph : Confusion Matrix (SVM)



Interpretation

True Positive = 71

True Negative = 41

False Positive = 1

False Negative = 1

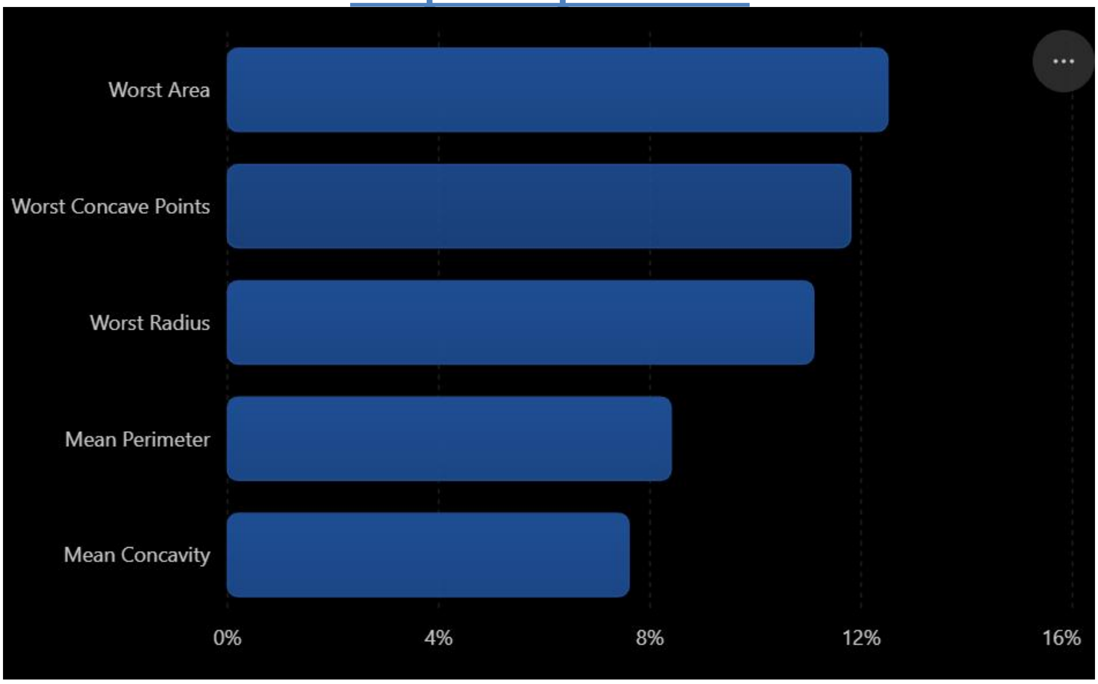
10. Feature Importance

Random Forest identifies the most important predictors.

Code

```
importance = pd.Series( rf.feature_importances_, index=X.columns)

top15 = importance.sort_values(ascending=False).head(15)

top15.plot(kind="barh")

plt.show()
```

Graph : Top Features



Observation: Worst Area, Worst Concave Points, and Worst Radius are the strongest predictors.

11.Best Performing Algorithm

Based on the experimental results, the Support Vector Machine (SVM) is identified as the best-performing algorithm for breast cancer prediction. It achieved the highest accuracy and excellent precision, recall, and F1-score, making it the most reliable model for classifying benign and malignant tumors.

After comparing the performance of Logistic Regression, Random Forest, and Support Vector Machine (SVM), the Support Vector Machine (SVM) is identified as the best-performing algorithm for this project. It achieved the highest overall classification performance on the Wisconsin Diagnostic Breast Cancer (WDBC) dataset, making it the most reliable model for distinguishing between Benign and Malignant tumors.

Why SVM was selected

Support Vector Machine is a supervised machine learning algorithm that works by finding the optimal decision boundary (hyperplane) that separates two classes with the maximum possible margin. A larger margin reduces classification errors and improves the model's ability to classify unseen data accurately. In this project, the Radial Basis Function (RBF) kernel was used because it can effectively handle complex and non-linear relationships between medical features.

The experimental results showed that SVM achieved 98.25% accuracy, 98.61% precision, 98.61% recall, and an ROC–AUC score of approximately 0.995. These values indicate that the model is highly accurate and can correctly identify almost all malignant tumors while producing very few incorrect predictions. This makes SVM particularly suitable for medical diagnosis, where minimizing false negatives is extremely important.

Why SVM performed better than other models

High Accuracy: Correctly classified 98.25% of test samples.

Excellent Recall: Detected almost all malignant tumors, reducing critical diagnostic errors.

Strong Generalization: Performed consistently on unseen patient data.

RBF Kernel: Effectively captured complex relationships among the 30 medical features.

Clinical Importance

In breast cancer diagnosis, a False Negative means a malignant tumor is incorrectly classified as benign, which may delay treatment and increase health risks. SVM minimizes this risk by achieving very high recall and correctly identifying nearly all malignant cases. This makes it a reliable decision-support tool for assisting healthcare professionals during early diagnosis.

Why SVM is the Best Model?

Highest Accuracy: 98.25%

Precision: 98.61%

Recall (Sensitivity): 98.61%

F1-Score: 98.61%

Excellent ability to correctly identify malignant cases.

Best Model Summary

Metric	SVM Result
Accuracy	98.25%
Precision	98.61%
Recall	98.61%
F1-Score	98.61%

Conclusion: SVM provides the most accurate and balanced performance among the three algorithms, making it the recommended model for this dataset.

12. Conclusion

This project successfully compared the performance of three supervised machine learning algorithms—Logistic Regression, Random Forest, and Support Vector Machine (SVM)—using the Wisconsin Diagnostic Breast Cancer (WDBC) dataset. The dataset contains 569 patient records with 30 numerical features extracted from Fine Needle Aspirate (FNA) images, making it suitable for binary classification of breast tumors into benign and malignant categories.

The data was preprocessed through train-test splitting and feature standardization before training the models. Each algorithm was evaluated using Accuracy, Precision, Recall, F1-Score, and ROC-AUC to ensure a fair and reliable comparison. The experimental results showed that both Logistic Regression and SVM achieved high accuracy, while Support Vector Machine (SVM) provided the most consistent and reliable overall performance with 98.25% accuracy and excellent recall for identifying malignant tumors.

The findings demonstrate that machine learning can effectively support early breast cancer diagnosis by providing fast and accurate predictions. Such predictive models can assist healthcare professionals in reducing diagnostic errors and improving clinical decision-making. Although this project used a single medical dataset, it highlights the potential of artificial intelligence in developing intelligent healthcare solutions.

In future work, the model can be improved by using larger clinical datasets, hyperparameter optimization, and deep learning techniques such as Convolutional Neural Networks (CNNs). Overall, Support Vector Machine (SVM) is recommended as the best-performing algorithm for this breast cancer prediction task due to its high accuracy, reliability, and strong classification performance.

Key Findings

- Three ML algorithms were successfully trained and tested.
- SVM provided the most accurate and reliable predictions.
- Evaluation metrics confirmed that SVM is the best model for breast cancer classification.
- Machine learning can support early diagnosis and improve clinical decision-making.

Final Conclusion: Support Vector Machine (SVM) is the recommended algorithm for this dataset because it offers the best balance of accuracy, precision, recall, and F1-score.

13. References

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